

PPE compliance detection on construction sites

Fine-tuning YOLOv8 to flag missing hardhats and hi-vis vests in site photography

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Repository: github.com/solomon8909/maicen0526-m4u3-ppe-detection

1 · Problem and success criteria

On an active industrial build such as the DDIP grain processing facility in Djibouti, PPE compliance is verified by walking the site. A safety officer covers one area at a time and the record is a paper form. We asked a narrow question: can a small object detector, trained on public data, flag a worker missing a hardhat or a high-visibility vest in a site photograph? The intended use is decision support — the model raises a flag, a person decides — not enforcement and not clearance to enter a hazardous area.

Three criteria were fixed before training: recall on **NO-Hardhat** of at least 0.70, because a missed violation is the costly error; overall mAP50 of at least 0.60, a realistic target for a 30-epoch nano model; and reproduction in the cloud with no credentials of any kind.

2 · Data and governance

The dataset is Construction Site Safety from Roboflow Universe, licensed CC BY 4.0. Five of its twenty-five classes were kept — Hardhat, NO-Hardhat, Safety Vest, NO-Safety Vest and Person — split 80 / 20 into 574 training and 143 validation images, resized to 640 × 640 with no augmentation. Masks, cones, plant and vehicles were dropped because they answer a different question from the one we set. The frozen export is published as a Release asset with the checksum **979ebeda...83c926**.

The pair of negative classes deserves a word. A detector only finds what is present, so it cannot directly report that a hardhat is absent. Labelling the bare head and the uncovered torso turns a missing item into something visible that the model can learn — at the cost of two classes that are intrinsically harder, because they depend on noticing what is not there.

Rights. Republishing the dataset is redistribution, so the frozen copy carries the upstream CC BY 4.0 licence and its attribution. No client or employer photography appears in the repository or its release, and none was uploaded to Roboflow, whose free plan publishes to Universe by default. The training images show real workers: a licence to reuse an image is not consent from the people in it, so the model is used for detection only, with no face recognition and no published face crops. The training library, Ultralytics YOLOv8, is AGPL-3.0 — acceptable for open academic work, but a licensing decision before any commercial deployment.

3 · Method and reproducibility

Roboflow was used to fork, version and export the dataset, and nothing more. The exact export we trained on was frozen, fingerprinted with a SHA256 checksum and published as a GitHub Release asset, and that release is what the notebook reads. A key-dependent download would have made reproduction depend on the reader holding an account and on Roboflow still serving that version later; a checksummed public file depends on neither. If the checksum does not match, the run stops rather than training on the wrong data.

Training: **yolo**v8n starting from COCO-pretrained weights, 30 epochs, batch 16, 640 px, seed 42, deterministic, on a free Colab T4 — 5.4 minutes. Evaluation on the validation split at confidence 0.25 and IoU 0.50, with ultralytics 8.4.163 and torch 2.11.0. Software versions, hardware, runtime and the dataset checksum are recorded automatically by the notebook and reproduced in the repository README.

4 · Results

Class	Precision	Recall	mAP50	mAP50-95
all	0.806	0.584	0.662	0.411
Hardhat	0.952	0.633	0.760	0.486
NO-Hardhat	0.700	0.510	0.545	0.291
NO-Safety Vest	0.767	0.499	0.573	0.338
Person	0.832	0.626	0.716	0.485
Safety Vest	0.779	0.653	0.717	0.456

Overall mAP50 of 0.662 clears the 0.60 we set. Recall on **NO-Hardhat** is 0.510, well short of the 0.70 we said we needed, and we report that as a failure against our own criterion rather than adjusting the criterion afterwards. Precision exceeds recall on every class (0.806 against 0.584 overall): the model is cautious, and the two classes carrying the safety signal — **NO-**

Hardhat and **NO-Safety Vest** — are its weakest on every measure. That is the failure direction our risk note warned about before we had numbers. The training curves add a caveat in the model's favour: validation mAP, recall and both losses were still improving at epoch 30, so the run stopped while the model was still learning. **Safety Vest** scores well but rests on only 49 validation instances and should not be leaned on.

5 · Error analysis

Every validation image was matched prediction-to-label by class at IoU 0.50: 128 false positives and 251 false negatives. Reading the worst cases image by image changed what those counts mean.

False positives. Four of the six most confident come from a single photograph of a retail queue, where the model called bare heads **NO-Hardhat** at 0.85–0.95. Every one of those calls is correct: the annotator labelled the front of the queue and stopped. A fifth is a pedestrian torso on a shopping street, unlabelled for the same reason. Our measured precision is therefore an underestimate. The sixth is a genuine error — a plain lime t-shirt on an excavator operator called **Safety Vest** at 0.84, which our class rules exclude because it carries no reflective banding.

False negatives. The six largest misses each fill 20–40% of their image, which rules out the usual small-object explanation. One is a worker behind scaffolding poles: the model found his vest at 0.83 but could not assemble the body across the pipes. One is a navy jacket with yellow-and-silver reflective banding that the model declined to call a vest, though it found the hardhat above it at 0.98. The rest are rear-view pedestrians and duplicated frames. Where objects genuinely are small, such as an aerial view of a crane, the distant workers are detected at confidences of 0.25–0.68: faint, not invisible.

Colour, not banding. Those two cases are one behaviour seen from both sides. The model calls a plain lime t-shirt a safety vest and refuses to call a banded navy jacket one, because almost every vest in training is fluorescent. It has learned hue as the signature of the class, and that is a data problem with a testable fix.

The finding underneath. A substantial share of the dataset is not construction imagery at all — shopping crowds, indoor webcam selfies, promotional and stock photography — labelled **NO-Hardhat** and **NO-Safety Vest** because the people in them wear ordinary clothes. The model has partly learned ‘bare head anywhere’ rather than ‘bare head on a site’, and the headline figure measures two domains at once. A sample review of the labels is documented in [docs/label_review.md](#).

The two error types are not equal for this use. A false alarm costs a safety officer a minute to check, and only becomes a problem in volume, when alert fatigue sets in. A false negative leaves a genuine hazard unflagged, and if the system earns trust it can lower vigilance below where it stood with no system at all. The model is therefore tuned to favour recall on the negative classes, with the alarm rate monitored rather than optimised.

6 · Limitations and next steps

- Not a safety system: it must never be the sole check before entry, and must never clear a worker as compliant.
- Not validated on our own sites. The repository is public and project photography is not redistributable, so both training and test images are public ones. Testing on real site images, in a private workspace, is the first step of any pilot.
- Weakest on occluded and crowded scenes rather than on small objects, and untested in night work, heavy dust or rain and in drone or crane-camera views.
- Two PPE items only — silent on harnesses, gloves, eye protection and boots, including fall protection, where the gravest construction injuries occur.
- Metrics come from a single validation split of 143 images, and **Safety Vest** from only 49 instances: estimates, not guarantees.
- The training data mixes construction and everyday imagery, so the reported figures are not a clean estimate of site performance in either direction.

Next data improvements, in priority order. First, break the colour bias: add non-fluorescent high-visibility workwear and bright non-workwear, and settle in the class definitions whether banding or hue decides the class. Second, filter the dataset to construction contexts and re-evaluate — a deletion rather than a collection exercise, after which the confident pedestrian false positives should disappear and the score should finally measure site performance. We expect the headline number to fall, and would regard that as the measurement becoming honest. Third, add roughly 200 occluded, crowded and rear-view site images, targeting the scaffolding and crowd misses that account for most of the 83 missed **Person** instances; the check is **Person** recall above 0.75. Longer training would also help — the curves show the run stopped early — but without better data a bigger model mostly learns the existing gaps more confidently.

AI use declaration: Generative AI (Anthropic Claude) was used to help structure the repository, draft notebook code and documentation, and review the brief against the rubric. All training runs, results, error interpretation and final wording were produced and checked by the group.